

Towards FAIR Documentation of Hypothetical 3D Reconstructions: The Case of C.-N. Ledoux's Unbuilt Projects

Fabrizio I. Apollonio¹, Federico Fallavollita¹ and Riccardo Foschi¹

¹ Department of Architecture, University of Bologna, Italy

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Abstract

1. Research problem

How can hypothetical 3D reconstructions of architectural heritage be documented in a transparent, reproducible, and FAIR-compliant way? Although digital reconstruction has become increasingly widespread, many 3D models remain difficult to evaluate, compare, and reuse because of insufficient documentation and the lack of shared methodological standards. This work addresses this gap by proposing a structured documentation framework aligned with the London Charter (2009), the Seville Principles (Lopez-Menchero, & Grande, 2011), the Critical Digital Model (Apollonio, Fallavollita, & Foschi, 2021), the Scientific Reference Model (Kuroczyński, Bajena, & Cazzaro, 2024), recent documentation practices, and the FAIR principles (Wilkinson, Dumontier, Aalbersberg, et al., 2016).

2. Scope

Hypothetical architectural reconstruction refers to the digital reconstruction of architectural artefacts that are either entirely lost or never realised, where parts of the geometry necessarily derive from scholarly interpretation rather than direct physical evidence. Unlike existing evidence-based workflows, the proposed methodology integrates semantic organisation, structured documentation, uncertainty communication, and FAIR publication into a single end-to-end process. The framework has been developed through the Erasmus+ CoVHer project (<https://covher.eu>) and validated over several years within a Problem-Based Learning course at the University of Bologna. It is demonstrated through case studies of Claude-Nicolas Ledoux's visionary unbuilt architecture.

3. Workflow

The methodology covers the entire reconstruction lifecycle through nine interconnected phases: (1) data acquisition; (2) critical evaluation of historical and architectural sources; (3) geometric analysis and 2D redrawing; (4) vectorisation; (5) semantically structured 3D

modelling; (6) visualisation; (7) uncertainty assessment and communication; (8) documentation; and (9) publication with emphasis on interoperability and accessibility.

4. Main methodological contributions

The framework is centred on two complementary documentation outputs. First, semantic segmentation structures architectural knowledge by classifying building components into a hierarchical system linked to their documentary evidence, improving documentation and interoperability through a semantic diagram (De Luca, Florenzano, & Veron, 2007; Apollonio, Gaiani, & Corsi, 2010). Second, uncertainty is evaluated using explicit criteria based on the availability, typology, authorship, quality, and Level of Detail of documentary sources (Foschi, Fallavollita, & Apollonio, 2024). The results are communicated through an uncertainty diagram that visualises the reliability of each reconstructed element. Together, these diagrams provide transparent, reproducible, and comparable documentation of hypothetical reconstructions.

5. Implementation

The modelling workflow combines continuous (NURBS) and discrete (mesh) representations to support accurate geometric modelling and analytical operations. Two free plug-ins developed assist researchers in verifying the geometry and calculating and visualising uncertainty. Artificial intelligence is employed selectively for specific architectural components within a fully documented workflow. The resulting models are published through the CoVHer platform (<https://repository.covher.eu/>), promoting accessibility, interoperability, and long-term reuse. The platform is represented in Figure 1.

6. Conclusion

The proposed workflow establishes a reproducible methodology for documenting hypothetical architectural reconstructions through the integration of structured documentation, semantic modelling, uncertainty assessment, and open publication. By producing semantic and uncertainty diagrams as mandatory documentation outputs, it improves transparency, interoperability, comparability, and long-term reusability of hypothetical architectural reconstructions, aligning with the FAIR and TRUST principles (Lin, Crabtree, Dillo, et al., 2020) for 3D data addressed by the FAIR 3D Heritage conference and by contemporary digital heritage infrastructures.

Although validated on unbuilt architectural case studies, further evaluation across additional architectural typologies and reconstruction contexts will be necessary to assess its broader applicability.

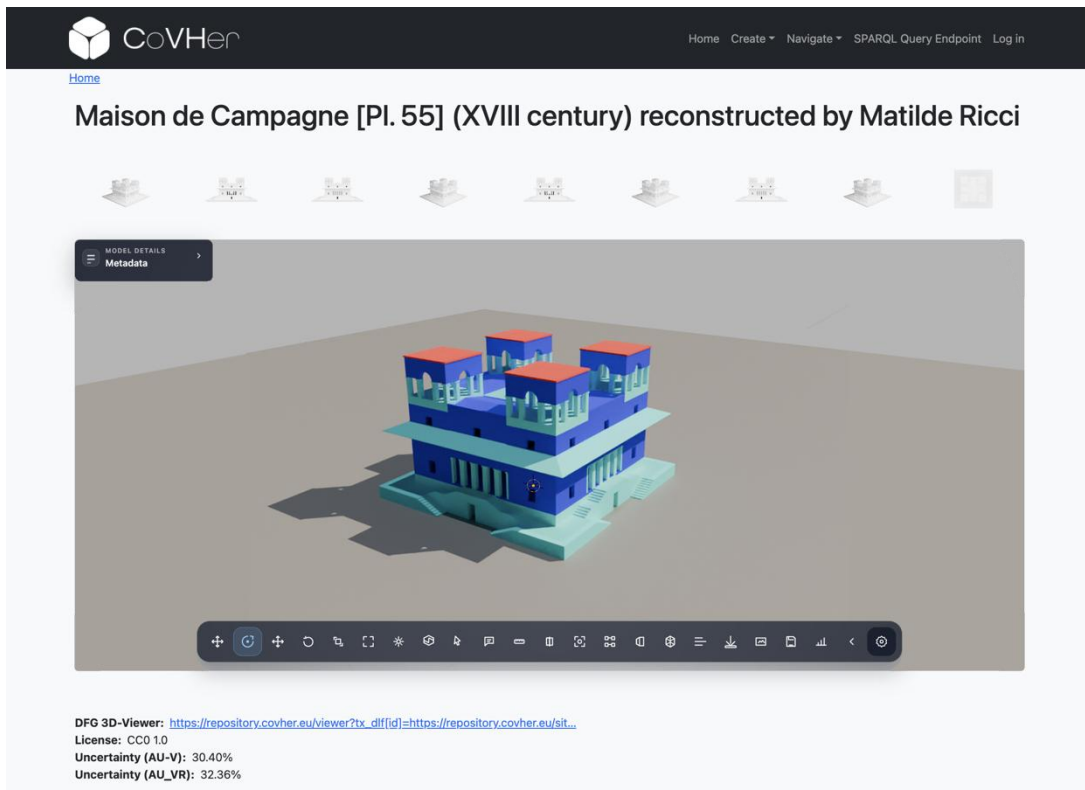


Fig. 1 Screenshot from the CoVHer platform, displaying the level of uncertainty of the hypothetical reconstruction of a Maison de Campagne (Pl 55) by C.-N. Ledoux.

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